SIH 2026 • SOFTWARE PROTOTYPE TRACK

AI-Assisted Diabetic Retinopathy Screening

Final Professional Architecture & Execution Plan — Locked to the Approved Decision Flow

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This document is the single authoritative specification for the project. It replaces all earlier drafts, including the version that included an image-enhancement model. The architecture below is fixed by one artifact only: the approved decision-flow diagram. Every section that follows is derived directly from that diagram and does not introduce logic the diagram does not contain.

Decision	Locked Answer
Number of AI models	Exactly 2 — Model 1 (Image Quality Check), Model 2 (DR Classification)
Enhancement model	Not part of the architecture. Removed permanently.
Borderline images	Reassessment → re-checked → if still unreliable, recapture/human review
Bad images	Recapture → re-enters the Image Quality Check
Good images	Proceed directly, without enhancement, to DR Classification
Uncertainty handling	Confidence/Uncertainty check after DR Classification, before the result is finalized
Explainability	Grad-CAM applied to the DR Classification output, not to the quality gate
Final safety layer	Human review, invoked only when required — not on every case
Positioning	AI-assisted screening support. Not an autonomous diagnostic device.

Prepared for SIH 2026. This is a research/prototype architecture. It is not a clinical deployment approval and must not be represented as one at any stage of the competition or in any demo material.

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1. Problem Statement

Automated diabetic retinopathy (DR) screening is only as trustworthy as the image handed to the classifier. In field conditions — camp screenings, rural PHCs, community vans — a meaningful fraction of captured fundus photographs are blurred, poorly illuminated, low-contrast, partially out of field, or otherwise not gradable. A classifier that is forced to output a severity grade for every image regardless of quality will still produce a confident-looking number on an unreliable input. That is the actual failure mode we are designing against — not "the model is inaccurate," but "the model was asked a question the input could not answer."

The brutal version of this problem: most student DR-screening projects fail at the demo stage because they show a single classifier accepting any image and printing a grade. That is not a screening system, it is a demo of a classifier. A real screening system has to know when to say *"I cannot answer this from what you gave me."* This project is built around that one capability, and everything else — the second model, Grad-CAM, human review — exists to support it.

Problem components

• Image-quality uncertainty is a distinct, earlier problem than disease-classification uncertainty, and must be resolved first.

• Poor images must never be silently converted into a forced DR prediction.

• A resource-limited screening setting needs an explicit, bounded failure-and-escalation path — not an infinite retry loop.

• The system must stay useful without ever claiming to replace an ophthalmologist's judgement.

• Validation must report both disease-classification performance and operational safety metrics (how often the system abstains, recaptures, or escalates), not accuracy alone.

Published evidence supports treating this as a first-class problem rather than an afterthought: a systematic review of AI-based DR grading found variable specificity across systems and inconsistent handling and reporting of ungradable images, concluding that evaluation in the intended screening pathway is essential before any implementation claim is made.[1] Real-world Indian evaluations additionally show that performance shifts across camera hardware and capture settings — which is exactly the variability a quality gate is meant to absorb before it reaches the disease classifier.[2]

2. Proposed Solution

A two-model, human-in-the-loop pipeline in which the first model's only job is to decide whether an image is trustworthy, and the second model's only job is to grade DR severity on images that have already been judged trustworthy. There is no enhancement model. Poor images are not algorithmically "fixed"; they are re-assessed and, where necessary, recaptured. This is a deliberate simplification from an earlier draft that included an enhancement stage — enhancement was cut because it adds a model whose failure mode (silently manufacturing plausible-looking detail on a bad image) is worse than the problem it claims to solve, and because it cannot be justified without a dedicated ground-truth study we do not have time to run in an SIH timeline.

Model 1 — Image Quality Assessment

Three-way classifier: Good / Borderline / Bad. Its output is a statement about image usability, not about disease. Only images that resolve to "Good" (directly, or via a successful reassessment/recapture) are permitted to reach Model 2.

Model 2 — DR Classification

Runs only on images that have cleared the quality gate. Produces a 5-class DR severity grade (No DR, Mild, Moderate, Severe, Proliferative). Its output is screening evidence for a human decision, not an autonomous diagnosis.

Reassessment / Recapture

Borderline images go through reassessment — a repeat quality evaluation, optionally after asking the operator for a new capture. Bad images go straight to recapture. In both cases the image is sent back through the Image Quality Check; there is no shortcut into Model 2. If reassessment still cannot certify the image as reliable, the case is routed to recapture/human review rather than looped indefinitely.

Human Review

Two distinct entry points feed human review: (a) images that remain unreliable after reassessment, and (b) DR predictions flagged as low-confidence or clinically high-risk after Model 2. Human review is the terminal safety layer for both problems — bad data and uncertain predictions — and it is not invoked for every case, only flagged ones.

3. Exact Flowchart Interpretation (Node-by-Node)

This section is a literal, node-by-node reading of the approved diagram. Nothing here is inferred beyond what the diagram states plus the one routing rule needed to make it operable: an image can only reach DR Classification once it has been certified as reliable by the Image Quality Check — directly, or after reassessment/recapture.

#	Node	Meaning / Transition
1	FUNDUS IMAGE	Raw retinal photograph enters the system.
2	IMAGE QUALITY CHECK	Model 1 runs. Output is one of three branches: Good, Borderline, Bad.
3a	GOOD	Image is usable as-is. Proceeds toward Reliable Original Image with no further gate.
3b	BORDERLINE	Image is marginal. Enters Reassessment before any decision is made.
3c	BAD	Image is unusable. Enters Recapture; the new capture re-enters the Image Quality Check (loop back to node 2).
4	REASSESSMENT	The Borderline image is re-evaluated (repeat quality scoring, and/or a fresh capture is requested).
5	STILL UNRELIABLE?	Decision point after reassessment. Two outcomes: No / Yes.
5-No	→ joins the Good path	Reassessment succeeded. The image is treated exactly like a Good image from this point on.
5-Yes	→ RECAPTURE / HUMAN REVIEW	Reassessment failed. The image is sent for a fresh capture and/or flagged for a human to inspect — it does not proceed to DR Classification on this pass.
6	RELIABLE ORIGINAL IMAGE	Convergence point. Every accepted path (Good directly, or Borderline-cleared-by-reassessment) arrives here holding the ORIGINAL, unmodified pixel data — confirming no enhancement step ever runs.
7	DR CLASSIFICATION	Model 2 runs on the reliable image and outputs a 5-class severity grade.
8	CONFIDENCE / UNCERTAINTY	Model 2's prediction is scored for confidence. Low-confidence or high-risk-grade predictions are marked for human attention.
9	GRAD-CAM	A visual attention map is generated for the DR prediction, regardless of confidence outcome, to support interpretation.
10	SCREENING RESULT	The grade, confidence flag, and Grad-CAM overlay are compiled into a single screening record.
11	HUMAN REVIEW WHEN REQUIRED	Terminal step. Only flagged records (from node 5-Yes or node 8) are routed to a human. Unflagged records are finalized as-is.

Non-negotiable routing rule, stated explicitly because it is the one place teams accidentally cheat under demo time pressure: ***no Borderline or Bad image may be handed to Model 2 directly.*** *Every accepted image reaches "Reliable Original Image" by passing through the Image Quality Check — either immediately (Good) or after a successful reassessment (Borderline → No). If your demo code has a path where a Bad image gets a DR grade, the architecture has been violated and the demo is not representative of the plan.*

4. Two-Model Architecture

Two models, two questions, two datasets, two evaluation protocols. Do not blend their training or their metrics — they are answering different questions and conflating them is the fastest way to produce numbers that look good and mean nothing.

	Model 1 — Image Quality	Model 2 — DR Classification
Question answered	Can we trust this image?	What DR severity does this trustworthy image show?
Output	Good / Borderline / Bad (3-class)	No DR / Mild / Moderate / Severe / Proliferative (5-class, grades 0–4)
Input	Any captured image, including poor ones	Only images certified reliable by Model 1
Architecture	Lightweight CNN (e.g. EfficientNet-B0 / MobileNetV3 backbone), fast inference for field use	Deeper transfer-learning CNN (e.g. EfficientNet-B3/B4 or ResNet50), optimized for grading accuracy over speed
Training signal	Quality-labeled dataset (blur, illumination, contrast, field-of-view, artifact labels)	APTOS 2019 severity labels (0–4)
Failure mode if wrong	A bad image is treated as usable → unreliable grade reaches the patient	A usable image is graded incorrectly → wrong severity reaches the patient
Mitigation for that failure	Reassessment/recapture loop; conservative thresholding (bias toward Borderline over Good when uncertain)	Confidence/uncertainty check; Grad-CAM for interpretability; human review on flagged cases

Engineering rule, stated bluntly: do not add a third model unless you can name the exact failure it fixes, the exact dataset it needs, and the exact metric that will prove it helped. "It would make the pipeline more impressive" is not a justification a judge will accept, and it is not one we accept internally either.

5. Reassessment / Recapture Logic

This is the mechanism that keeps unreliable images out of Model 2 without simply throwing them away. It has exactly two entry points and one exit rule.

Trigger	Action	Exit condition
Model 1 output = Borderline	Reassessment: re-run Model 1 on the same frame with a stricter decision threshold, and/or prompt the operator for one repeat capture	Still unreliable? = No → join Good path. = Yes → Recapture / Human Review
Model 1 output = Bad	Recapture: reject immediately, return a specific reason code (blur / dark / glare / off-field / low-contrast) to the operator, request a new capture	New capture re-enters the Image Quality Check from node 2. No cap-then-forward shortcut.
Recapture/Human Review reached (from node 5-Yes)	Flag the case; a human decides whether to attempt one more capture on-site or refer the patient for in-person examination	Human decision is logged as the terminal outcome for this image; the case does not silently disappear

Retry limit: a hard cap of **2 recapture attempts per patient per session** is enforced in the reference implementation. This is a product decision, not a modeling one — it exists so that a patient with a genuinely difficult eye (small pupil, cataract, uncooperative gaze) is escalated to a human quickly instead of being cycled through the same automated loop. The cap must be reported in the results as a safety parameter, and the rate at which it is hit must be measured (see Section 13).

6. DR Classification (Model 2)

Model 2 is a transfer-learning image classifier trained on APTOS 2019, predicting the standard International Clinical DR Severity Scale collapsed to 5 grades: 0 (No DR), 1 (Mild NPDR), 2 (Moderate NPDR), 3 (Severe NPDR), 4 (Proliferative DR). Referable DR is defined at grade ≥ 2, matching the team's existing planning material and standard screening practice.

Design choices and their reasons

• Backbone: EfficientNet-B3 (or ResNet50 as a fallback if compute or time is constrained), pretrained on ImageNet, fine-tuned end-to-end on APTOS 2019 with class-weighted loss — APTOS is heavily skewed toward grade 0, and an unweighted loss will collapse the model toward always predicting “No DR.”

• Input images are resized and normalized using fixed preprocessing (circular fundus crop, contrast-limited adaptive histogram equalization is applied only as a training-time augmentation, never as a silent runtime enhancement that would contradict the no-enhancement decision).

• Loss: focal loss or class-weighted cross-entropy, chosen empirically — report both and pick the one with better minority-class (Severe, Proliferative) recall, not the one with the higher raw accuracy.

• Output is a full 5-class probability vector, not just an argmax label — the probability vector is what feeds the Confidence/Uncertainty step in Section 7.

Brutal note on accuracy claims: with APTOS alone and a competition timeline, do not promise clinical-grade sensitivity on Severe/Proliferative DR. These are minority classes with a few hundred images total. State the honest, measured sensitivity per class in the results, and let the confidence/human-review layer compensate for the classes where the model is weakest — that is the actual point of having those layers.

7. Confidence and Uncertainty Handling

This step sits between DR Classification and Grad-CAM in the diagram, and its job is to decide whether Model 2's output can be trusted enough to finalize without a human, or must be flagged.

Signal	How it is computed	Action if it fires
Low softmax confidence	Top-1 predicted-class probability below a tuned threshold (start at 0.60, tune on validation set)	Flag for human review
Class ambiguity	Difference between top-1 and top-2 class probabilities below a margin threshold (e.g. < 0.15)	Flag for human review
Predictive uncertainty (optional, stretch goal)	Monte-Carlo Dropout or a 5-model ensemble; variance of predictions across stochastic passes	High variance → flag for human review
High-risk grade	Predicted grade is Severe or Proliferative, regardless of confidence	Always flag for human review — confidence is not allowed to suppress a high-severity flag

This is deliberately conservative: a high-risk grade is flagged even if the model is confident, because the clinical cost of a missed Severe/Proliferative case is far higher than the cost of one extra human review. Confident low-risk predictions (grade 0–1 with high margin) are the only class of case allowed to skip straight to a finalized Screening Result without a human touching it.

8. Grad-CAM Explainability

Grad-CAM is applied to Model 2 only, after the Confidence/Uncertainty step, and it runs on every case that reaches this node — flagged or not — because a human reviewer needs the same visual evidence whether or not the case was auto-flagged, and because it is part of the audit trail attached to every screening result.

• Method: Grad-CAM (or Grad-CAM++ for sharper localization) computed on the final convolutional layer of Model 2, overlaid on the original fundus image as a heatmap.

• Claim boundary, stated explicitly to reviewers and judges: Grad-CAM shows which pixels most influenced the model's decision. It does not prove that a lesion exists at that location, and it is not a lesion-segmentation output. Any slide or report that says “Grad-CAM detects microaneurysms” is overstating what the method does.

• Where IDRiD lesion annotations are available, Grad-CAM regions can be qualitatively compared against annotated lesion locations for a small representative set of cases — this is a sanity check, not a validated correspondence metric, and should be reported as such.

• Grad-CAM output is stored alongside the predicted grade and confidence flag in the screening record, so a human reviewer sees the same three things: grade, confidence status, and heatmap, in one view.

9. Human Review

Human review is the terminal node in the diagram, and it is reached from exactly two places, never from a general “review everything” policy — that would defeat the purpose of automation and is not what the diagram specifies.

Entry point	Who reviews	What they see	What they decide
Node 5-Yes: image still unreliable after reassessment	Trained field operator or supervising clinician	The rejected image, the reason code, and the recapture attempt count	Attempt one more capture on-site, or refer the patient to an in-person eye exam
Node 8: low-confidence / ambiguous / high-risk DR prediction	Ophthalmologist or trained medical professional (tele-review acceptable)	The reliable original image, predicted grade, confidence flag, and Grad-CAM overlay	Confirm, override, or downgrade the grade; decide on referral urgency

This split matters and should not be collapsed in the demo: an ungradable image is an *operator-level* problem (get a better photo), while an ambiguous DR grade is a *clinical* problem (get a better-trained eye on it). Routing both to “some human” without distinguishing the two wastes clinician time on recapture requests they cannot help with.

Multisite AI DR screening deployments that pair automated grading with specialist over-read of flagged cases report this combination as an effective safety mechanism at scale,[3] and telemedicine DR programs using reflex imaging with manual over-read report a similar pattern.■ Human review here follows that established model rather than inventing a new one.

10. Technology and Project Structure

Component	Technology	Responsibility
Language	Python 3.10+	All training, preprocessing, orchestration, evaluation
Deep learning	PyTorch	Model 1 and Model 2
Image processing	OpenCV, PIL	Loading, resizing, quality feature extraction (blur/illumination/contrast metrics)
Data / metrics	NumPy, Pandas, scikit-learn	Dataset handling, confusion matrices, ROC/AUC, class-wise reports
Explainability	pytorch-grad-cam	Grad-CAM heatmaps for Model 2
Compute	Kaggle Notebooks (free GPU)	Training and experiments for both models
Interface	Streamlit (or Flask, optional)	Demo UI: upload image → see routing decision → see result
Version control	Git / GitHub	Code, experiment logs, results, documentation

Repository structure

```
SIH2026-DRScreeningPipeline/
■■■ src/
■   ■■■ quality/           # Model 1: training, inference, thresholds
■   ■■■ classification/    # Model 2: training, inference, Grad-CAM
■   ■■■ pipeline/          # routing logic implementing Section 3 exactly
■■■ tests/
■   ■■■ unit/               # per-module tests
■   ■■■ integration/        # full-pipeline routing tests
■   ■■■ edge_cases/         # see Section 18
■■■ kaggle/                # notebook exports for both models
■■■ results/                # metrics, confusion matrices, Grad-CAM samples
■■■ docs/                   # this document and supporting notes
■■■ presentation/           # slides, diagrams, exported PDF
■■■ demo/                   # demo assets and Streamlit app
```

11. Dataset Strategy

Dataset	Used for	Notes
APTOS 2019 (Kaggle)	Model 2 training/validation — 5-class DR severity	~3,662 labeled images, class-imbalanced toward grade 0; primary dataset for the whole project
EyePACS / DeepDRiD (or in-house labeled set)	Model 1 training — quality labels (Good/Borderline/Bad)	Quality labels are not provided by APTOS itself; label a stratified subset manually or use a public quality-labeled set, with clear labeling criteria documented in docs/
IDRiD	Grad-CAM sanity-check against lesion annotations on a small sample	Not used to train either model; used only for qualitative explainability review
Messidor-2	Optional external evaluation of Model 2 if time permits	Cross-dataset performance drop is expected and should be reported honestly, not hidden

Labeling Model 1's training data is the single most time-consuming task in this plan and should be scheduled first, not last. A practical approach: synthetically degrade a subset of Good-quality APTOS/EyePACS images (Gaussian blur, brightness/contrast shifts, synthetic vignetting) to generate Borderline/Bad examples at controlled severity, combined with a smaller set of genuinely poor real-world captures if available. Document the exact degradation parameters used — a judge will ask.

12. Research Question and Hypothesis

Research question: Does a two-model, image-reliability-aware routing pipeline reduce unreliable/forced DR predictions and improve operational safety, compared with applying a DR classifier directly to every captured image, without materially reducing screening throughput?

Hypothesis: The proposed pipeline will show a measurably lower forced-prediction rate on ungradable images and a higher effective sensitivity on reliable images than the direct-classification baseline, at the cost of a bounded recapture/human-review overhead that stays below an agreed operational threshold (target: under 20% of total captured images requiring any escalation).

Arm	Flow	Purpose
Baseline	Image → Model 2 directly, no quality gate	Measures the cost of ignoring image quality — the honest comparison point
Proposed	Image → Model 1 → Reliable Original Image → Model 2 → Confidence check → Grad-CAM → Result	Measures the value of reliability-aware routing exactly as diagrammed
Proposed + human loop	Proposed, with the human-review terminal node active	Measures the fully operational, deployable version of the system

13. Evaluation Metrics

Metrics are grouped by what they actually test. Reporting only overall accuracy for Model 2 is not acceptable given the class imbalance in APTOS — a model that always predicts “No DR” scores roughly 50% accuracy and is clinically useless. Every metric below must be reported per-class where applicable.

Layer	Metrics	Why this metric, specifically
Model 1 (Quality)	Class-wise precision/recall, confusion matrix, Good→Bad and Bad→Good error rates	The two error rates are asymmetric in cost: letting a Bad image through is worse than rejecting a Good one, and the report must show both separately, not a single averaged score
Model 2 (DR grade)	Per-class sensitivity/recall, specificity, quadratic-weighted kappa, AUC-ROC (one-vs-rest), F1, full confusion matrix	Quadratic-weighted kappa is the standard APTOS/DR-grading metric because it penalizes a 2-grade miss more than a 1-grade miss — raw accuracy does not
Referable DR (grade ≥ 2)	Sensitivity, specificity, PPV/NPV at the chosen operating threshold	This is the number that matters for real referral decisions, independent of exact grade
Routing / safety	Forced-prediction rate (baseline only), recapture rate, repeated-failure rate (hit retry cap), human-review rate, average recaptures per accepted image	These are the metrics that justify the whole architecture; without them the project cannot claim the pipeline is safer, only that it is more complex
Explainability	Qualitative Grad-CAM review on a representative sample against IDRiD lesion locations; no quantitative claim beyond this	Prevents overstating what Grad-CAM proves (see Section 8)
Generalization (optional)	Model 2 performance on Messidor-2 vs. APTOS held-out set	Honest cross-dataset drop is expected and should be shown, not omitted

14. Real-Life Scope

The intended deployment context is AI-assisted triage in settings where large volumes of fundus images must be screened and specialist ophthalmology capacity is limited relative to demand — community screening camps, rural PHCs, and school/workplace vision programs run alongside diabetes care.

Scenario	How this architecture helps	What it does not solve
Community screening camps	Fast first-pass triage; only usable images consume the more expensive DR-grading step	Does not replace the need for a trained operator to physically position patients and cameras
Rural / resource-limited PHCs	Works with standard, non-specialist fundus cameras; quality gate catches hardware/lighting inconsistencies before they become misdiagnoses	Does not compensate for a camera that is fundamentally unsuitable (e.g. no pupil dilation capability where needed)
High-volume screening drives	Automates the repetitive first-pass read; escalates only the minority of genuinely uncertain or high-risk cases to scarce clinician time	Does not eliminate the need for a clinician entirely — referable and flagged cases still require one
Clinical decision support	Gives the reviewing clinician a pre-structured record: grade, confidence, heatmap, in one screen	Does not make the final referral decision — that remains a human responsibility by design

Recent Indian evidence on offline, AI-assisted community DR screening reports genuine practical value but also notes that outcomes remain sensitive to image quality — which is precisely the argument for keeping the quality gate as the first, non-optional stage of this pipeline rather than a nice-to-have.■

15. Safety and Limitations

Stated bluntly, so nobody on the team accidentally oversells this in a pitch: here is exactly what can and cannot be claimed.

Can claim	Cannot claim
The system can classify image usability with measured precision/recall	Quality classification guarantees downstream clinical safety
Model 2 predicts DR severity on data resembling its training/validation distribution, with reported per-class performance	The model autonomously diagnoses patients or replaces an eye exam
Grad-CAM visualizes regions that influenced the prediction	Grad-CAM proves a specific lesion (microaneurysm, hemorrhage, etc.) is present
External-dataset (Messidor-2) testing indicates generalization behavior	External testing is equivalent to clinical validation
Human review provides final adjudication on flagged cases	The presence of AI removes the need for qualified clinical judgement anywhere in the pathway
The architecture reduces forced predictions on poor images, measured against a stated baseline	The system has been validated for deployment in any specific clinical setting

This boundary is not just ethical hygiene — it is also what the literature says is required. Systematic reviews of AI DR grading systems flag variability and risk of bias across studies and explicitly call for evaluation within the intended screening pathway before any implementation claim.[1]¶ This project reports experiments and their limitations; it does not claim clinical validation at any point in the SIH materials, demo, or presentation.

16. Implementation Plan

Phase	Work	Definition of done
1. Freeze architecture	Lock this document as the single spec; remove enhancement references from all prior materials	Every team member can draw Section 3's flow from memory correctly
2. Data prep	Label/generate quality dataset (Good/Borderline/Bad); prepare APTOS splits (train/val/held-out test)	Class distributions and labeling criteria documented in docs/
3. Model 1	Train quality classifier; tune Good/Borderline/Bad thresholds	Confusion matrix + per-class precision/recall reported; Good→Bad and Bad→Good error rates specifically called out
4. Model 2	Train DR classifier on APTOS with class-weighted loss	Per-class sensitivity, quadratic-weighted kappa, AUC reported on held-out test set
5. Routing pipeline	Implement Section 3's flow exactly, including the retry cap from Section 5	Integration tests (Section 18) pass for every branch: Good, Borderline→No, Borderline→Yes, Bad, retry-cap-hit
6. Confidence layer	Implement thresholds from Section 7; wire high-risk-grade override	Flagging behavior verified against a hand-labeled set of ambiguous/high-risk predictions
7. Grad-CAM	Integrate pytorch-grad-cam on Model 2; overlay generation for every case reaching node 9	Representative Grad-CAM gallery produced, including on IDRiD comparison samples
8. Human review UI	Build the two-entry-point review screen described in Section 9	A reviewer can see reason/attempt count (recapture cases) or grade/confidence/heatmap (grading cases) and log a decision
9. Baseline comparison	Implement and run the direct-classification baseline from Section 12	Baseline vs. proposed metrics table completed
10. Freeze results	Finalize all metrics, limitations section, demo script, and presentation	Results, limitations, and demo are internally consistent with this document

17. Team Responsibilities

Assign these as named owners, not shared duties — shared ownership of a pipeline stage is how integration bugs go unnoticed until the night before the demo.

Role	Owns	Hands off to
Model 1 owner	Quality dataset, training, threshold tuning, Section 4/13 metrics for Model 1	Pipeline owner (inference API + thresholds)
Model 2 owner	APTOS training, class-weighting strategy, Section 6/13 metrics for Model 2	Pipeline owner + Grad-CAM owner
Grad-CAM / explainability owner	Grad-CAM integration, IDRiD comparison sample, claim-boundary language in all materials	Human review UI owner
Pipeline / routing owner	Section 3 flow implementation, retry-cap logic, confidence-layer wiring, integration tests	Human review UI owner; demo owner
Human review UI + demo owner	Review screen, Streamlit demo app, end-to-end walkthrough script	Presentation owner
Evaluation / metrics owner	Baseline implementation, all Section 13 metrics, results tables, honest limitations write-up	Presentation owner
Presentation / docs owner	This document's maintenance, slide deck, final PDF export, evidence-base citations	—

18. Testing and Edge Cases

The routing logic in Section 3 is the part most likely to be silently wrong under demo pressure. Test it explicitly, not just the two model accuracies in isolation.

• Good image → confirm it reaches Model 2 without ever touching the Reassessment/Recapture nodes.

• Borderline image, reassessment succeeds → confirm it joins the Good path and reaches Model 2 with the original (not enhanced) pixel data.

• Borderline image, reassessment fails → confirm it routes to Recapture/Human Review and does NOT reach Model 2 on this pass.

• Bad image → confirm it routes to Recapture and re-enters the Image Quality Check, never Model 2 directly.

• Retry-cap hit (2 recaptures exhausted) → confirm the case is force-escalated to human review rather than looping a third time.

• Confident, low-risk DR prediction (grade 0–1, high margin) → confirm it is finalized without a human-review flag.

• Low-confidence DR prediction → confirm it is flagged regardless of predicted grade.

• High-risk grade (Severe/Proliferative) at high confidence → confirm it is still flagged — this is the case most likely to be coded incorrectly, since “high confidence” intuitively suggests no flag is needed.

• Corrupt/unreadable file upload → confirm the system fails gracefully with a clear error, not a silent crash or a default grade.

• Non-fundus image (e.g. an unrelated photo) submitted → confirm Model 1 routes it toward Bad/rejection rather than Model 2 attempting a grade on nonsense input.

• Normal fundus images spanning all five DR grades → confirm each grade is reachable and correctly reported end-to-end.

• Out-of-distribution or unusual cases (where available) → confirm the system degrades to a flagged/uncertain state rather than a confident wrong answer.

19. Core vs Optional Scope

If time runs short before the SIH deadline, cut from the bottom of this list, not the top. Every core item is load-bearing for the claims made in Sections 1, 2, and 15.

Item	Status	Why
Model 1 (quality gate)	Core	The entire problem statement depends on this existing
Model 2 (DR classification)	Core	The disease-screening function of the system
Reassessment/Recapture routing, exactly as in Section 3	Core	This is the mechanism the project claims novelty for
Confidence/uncertainty flagging	Core	Required to make the human-review claim honest
Grad-CAM	Core	Explicitly required by the diagram and expected by SIH judges for any medical-imaging AI project
Human review UI (functional mock, not a full clinician portal)	Core	Needed to demonstrate the terminal node, not to be a production tool
Baseline (direct-classification) comparison	Core	Without it, Section 12's research question cannot be answered
Monte-Carlo Dropout / ensemble uncertainty	Optional	Improves rigor of the confidence signal but a softmax-threshold baseline is defensible on its own
Messidor-2 external evaluation	Optional	Strengthens the generalization discussion but is not required to prove the core architecture works
IDRiD quantitative lesion-correspondence scoring	Optional — and should stay qualitative even if attempted	A quantitative claim here risks overstating what Grad-CAM proves (Section 8)
DRIVE vessel segmentation	Out of scope	Not required by the approved architecture; do not add a third model without the justification test in Section 4
MATLAB reference implementation	Out of scope for execution	Retained only as architecture reference material, not the training path

20. Non-Destructive Processing Rule

This section exists because it is the rule most likely to be violated by accident, not by intent — someone adds a "quick contrast fix" during preprocessing and nobody notices it has quietly become a clinical enhancement step. State it once, explicitly, and enforce it in code review.

Rule: *no diagnostic image enhancement, reconstruction, denoising, sharpening, brightness manipulation, or contrast manipulation is ever applied to the original clinical image before a prediction is made on it. The image that reaches Model 2 is pixel-for-pixel the same retinal capture that reached Model 1, subject only to standard, non-clinical model-input operations.*

Allowed (model-input operations only)	Not allowed (clinical enhancement)
Resizing to the network's fixed input dimensions	CLAHE, histogram equalization, or any contrast stretch applied at inference time
Pixel-value normalization (mean/std scaling) required by the model	Denoising or deblurring filters applied to "rescue" a poor image
Standard color-space/format conversion (e.g. JPEG to RGB tensor)	Brightness/gamma correction applied to a dark image before grading it
Circular fundus-field cropping for consistent framing	Any generative or reconstructive model (super-resolution, inpainting) touching the image

Why this matters more than it might seem: an enhancement step that quietly improves a Borderline image's apparent quality can manufacture visual detail that was not actually captured, and a DR grade produced from manufactured detail is not a grade the team can defend under questioning. Reassessment (Section 5) replaces enhancement precisely because reassessment only ever re-evaluates or re-captures — it never modifies what the camera actually recorded. The "Reliable Original Image" label at node 6 of the flowchart (Section 3) is a factual claim, not a name, and it must remain true in the implementation.

21. Notebook Plan (N1–N4)

Work is organized into four Kaggle notebooks with clean boundaries, so two people can work in parallel without stepping on each other's code.

Notebook	Scope	Status
N1 — DR Classifier	Data loading, train/val/test split, transfer-learning model (Section 6), training, validation, testing, metrics, model export	Core
N2 — Image Quality	Quality feature extraction, Good/Borderline/Bad logic and threshold development, reassessment implementation, edge-case handling	Core
N3 — Explainability	Load the trained Model 2 checkpoint, generate predictions, produce Grad-CAM visualizations, assess qualitative lesion correspondence against IDRiD where annotations permit	Core
N4 — Vessel Segmentation	Optional DRIVE-based vessel segmentation	Optional — only start once N1–N3 are complete and stable

22. Milestones to the Internal Hackathon — 15 September 2026

Internal hackathon date: **15 September 2026**. As of this document, that leaves 8 working days. The schedule below assumes one day per major pipeline stage, with integration and freeze days at the end — it deliberately does not leave room for optional features, per the MVP split in Section 23.

Date	Milestone	Key deliverables
7 Sep — Day 1	Freeze scope & setup	Finalize this architecture; remove enhancement from all documents/code plans; create the GitHub repo structure (Section 10)
8 Sep — Day 2	Data + baseline DR setup	Download/organize APTOS 2019; EDA and class-distribution check; create train/validation/test splits
9 Sep — Day 3	DR model training (Model 2)	Train/fine-tune the classifier; record loss and metrics; save checkpoints; generate confusion matrix and per-class report
10 Sep — Day 4	Image quality gate (Model 1)	Implement blur, illumination, contrast, noise/artifact and field-of-view checks; define and tune Good/Borderline/Bad thresholds
11 Sep — Day 5	Reassessment + integration	Implement Borderline reassessment without pixel modification; connect the quality gate to the DR classifier exactly per Section 3's routing
12 Sep — Day 6	Confidence + Grad-CAM	Add confidence/uncertainty output (Section 7); implement Grad-CAM; generate clean example overlays across Grades 0–4
13 Sep — Day 7	A/B/C experiment	Run direct classification vs. quality-gated vs. quality-gated+reassessment (Section 12); produce core metrics and rejection/reassessment statistics
14 Sep — Day 8	Demo + presentation freeze	Freeze code; build the demo workflow (Section 24); finalize architecture diagram, results slides, limitations and safety language
15 Sep	INTERNAL HACKATHON	Present a stable end-to-end demo. Priority order: working pipeline → clear research question → measurable safety metrics → polish

23. Hackathon MVP: Must-Have vs Nice-to-Have

This is the same discipline as Section 19's core-vs-optional scope, sharpened to the 8-day hackathon clock specifically. If a feature is not in the left column, it does not get built before 15 September, no exceptions.

Must have by 15 Sep	Nice to have after the hackathon
Working APTOS-trained DR classifier	Messidor-2 external evaluation
Image quality gate (Model 1)	Advanced calibration (temperature scaling, Platt scaling)
Good/Borderline/Bad routing exactly per Section 3	More sophisticated uncertainty methods (MC-Dropout, ensembles)
Non-destructive reassessment (Section 20)	Automated, quantitative lesion-correspondence study
End-to-end integration (Model 1 → Model 2 → confidence → Grad-CAM)	DRIVE vessel segmentation (N4)
Confidence output	Edge/cloud production deployment
Grad-CAM examples across all 5 grades	Full web/mobile production UI
A/B/C comparison with core metrics	Additional architectures/ensembles
Demo, README, and presentation	Clinical prospective study

24. Demo Scenarios for the Hackathon

Prepare at least four polished, pre-selected demo cases — do not rely on live, unrehearsed uploads for a judged demo. Each case should visibly preserve the original image and make it obvious the system never states certainty when the input or the prediction is unreliable.

Case	Input	Expected on-screen behavior
1 — Good image	Clean, well-lit fundus photo	Passes the quality gate → DR grade predicted → confidence shown → Grad-CAM overlay displayed
2 — Bad image	Severely blurred / dark / off-field capture	Fails the quality gate → no DR grade is shown → specific recapture instruction is displayed (Section 25)
3 — Borderline image	Marginal quality capture	Enters reassessment → either proceeds to grading if it clears, or abstains and requests recapture if it does not
4 — Uncertain prediction	Good-quality image, but Model 2 confidence is low	Grade is shown but marked low-confidence → human-review flag is displayed, not a silent final answer

25. Suggested Screening Report Format

The screening record produced at node 10 (Section 3) should render consistently regardless of which path the image took. Two examples below: an accepted, graded case and a rejected, ungradable case.

```
Image Quality:      GOOD
Quality Status:      Reliable
DR Prediction:        Grade 2 — Moderate DR
Confidence:            87%
Explainability:         Grad-CAM generated
Action:                    Refer for ophthalmic evaluation / human review per workflow

Image Quality:      BAD
Reason:                   Insufficient field of view / severe blur / inadequate illumination
DR Prediction:           Not generated
Action:                    Recapture image. Do not display a DR grade for an image
                                  that fails the reliability gate.
```

The rule embedded in the second example is absolute and testable (see Section 18): a Bad or still-unreliable image must never carry a DR grade in its report, even a low-confidence one. “No grade” plus a recapture reason is the correct output, not “a grade with a warning.”

26. What Makes This Project Different

The core differentiator: this project treats image reliability as part of the medical-AI decision process itself, rather than assuming every input is trustworthy and only worrying about the disease-classification model.

• Quality-first screening — the very first decision the system makes is about the image, not the disease.

• Non-destructive retinal-image handling — no enhancement stage, no pixel manipulation before prediction (Section 20).

• Borderline-image reassessment, rather than a binary accept/reject on marginal captures.

• Explicit abstention for unreliable cases — the system is allowed to say it cannot answer.

• Confidence/uncertainty-aware routing on the disease prediction itself, not just on image quality.

• Grad-CAM explainability with an explicitly stated claim boundary (Section 8).

• A genuine human-in-the-loop escalation path with two distinct, purpose-specific entry points (Section 9).

• An A/B/C experimental design that actually measures whether the reliability-aware pipeline helps, instead of asserting that it does.

27. 30-Second Pitch

“Our project addresses a major weakness in automated diabetic-retinopathy screening: the AI may be confident even when the retinal image is not reliable. We therefore place an image-reliability gate before the DR model. The system evaluates blur, illumination, contrast, artifacts and retinal field of view, reassesses borderline images without altering the original clinical pixels, and abstains when the image remains unreliable. Only reliable images reach the five-class DR model. We then quantify confidence and use Grad-CAM to explain the prediction, with uncertain cases routed for human review. The goal is safer, more trustworthy and more practical AI-assisted retinal screening.”

28. Team Execution Priorities

In strict order. Do not start priority N+1 work before priority N is functionally working end-to-end, even if it means an individual owner appears idle for part of a day — a half-finished Grad-CAM integration is worthless if the DR classifier underneath it doesn't run reliably yet.

Priority 1: Get one reliable DR classifier (Model 2) running end-to-end.

Priority 2: Build a defensible quality gate (Model 1) and document its thresholds.

Priority 3: Integrate reassessment and abstention (Sections 3 and 5).

Priority 4: Add confidence scoring and Grad-CAM (Sections 7 and 8).

Priority 5: Run the A/B/C comparison (Section 12) and produce the core metrics table.

Priority 6: Freeze a polished demo and presentation on 14 September.

29. Evidence Base and References

Cited to support specific claims made in this document (image-quality variability, real-world performance drift, and human-over-read safety), not as a general literature dump.

1. Zhelev, Z. et al. Systematic review of AI-based diabetic retinopathy grading. Journal of Medical Screening, 2023 — supports the claims in Sections 1 and 15 on variable specificity and inconsistent handling of ungradable images.

2. Real-world AI DR screening evidence from primary care in India, 2026 — supports the claim in Section 1 on performance variability across camera hardware and settings.

3. Multisite deep-learning DR screening with specialist over-read, The Lancet Digital Health — supports the human-review model described in Section 9.

4. AI telemedicine DR screening with reflex imaging and manual over-read, American Journal of Ophthalmology — supports the human-review model described in Section 9.

5. Pragmatic AI-assisted DR screening study in India, 2026 — supports the real-life scope discussion in Section 14 on image-quality sensitivity in community screening.

6. Systematic review / meta-analysis of AI DR screening in real-world settings, 2024 — supports the safety and limitations claims in Section 15.

7. Retinal fundus image-quality assessment review, 2023 — background reference for the Model 1 design approach in Section 4.

8. Team source materials: TEAM_PLAN.md, the SIH 2026 project overview, and the SIH 2026 Full Project Blueprint (Trust-Aware and Explainable AI-Based DR Screening, internal hackathon 15 Sept 2026) — basis for the dataset-to-feature mapping in Section 11, the technology stack in Section 10, and the hackathon milestone plan, MVP split, demo scenarios, screening report format, pitch, and execution priorities in Sections 22–28.

This document is prepared for SIH 2026 as a research/prototype architecture plan. It is not a clinical deployment approval, has not undergone clinical validation, and must not be presented as either in any team material.